Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.Tech, Mechanical Engineering, III
Name of the Paper: Engineering Mechanics
Paper Code: TME-306
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. What is the principle of superposition of forces, and how does it describe the combined effect of multiple forces acting on a body? What is the principle of transmissibility of forces, and how does it explain the impact of changing the point of application of a force along its line of action on a rigid body? CO 1

OR

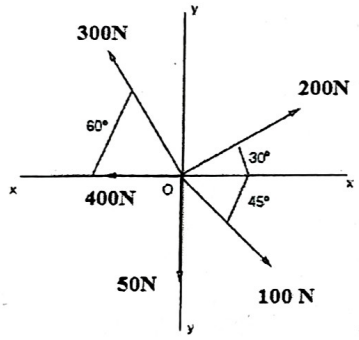
b. What are the different types of forces in mechanics, and how are they classified? and describe the characteristics of a force. CO 1

Q2. (10 Marks)

a. What is equilibrium in mechanics, what are the conditions required for an object to be in static equilibrium for different force system? CO 1

OR

b. Determine the resultant of the coplanar concurrent force system shown in figure. Compute the magnitude, sense, and the angle of inclination with the x-axis. CO 2

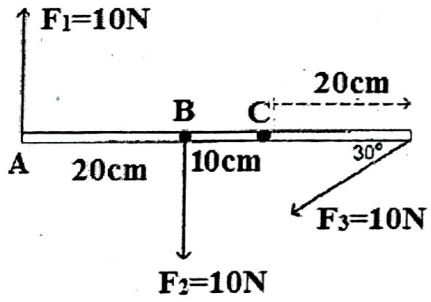


Q3. (10 Marks)

a. Explain resolution of force into a force and couple. CO 1

OR

b. Length of a rod is 50 cm. Three forces act on the rod, as shown in figure below. If the axis of rotation is point C, what is the net of the moment of force. CO 2



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Q4. (10 Marks)

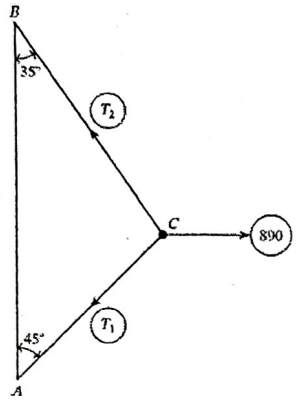
a. Find out the centroid of parabolic Spandrel($y=kx^2$) coordinate $x_1,y_1=(0,0)$ and $x_2,y_2=(a,b)$. CO 3

OR

b. Find out the centroid of semi-circle. CO 3

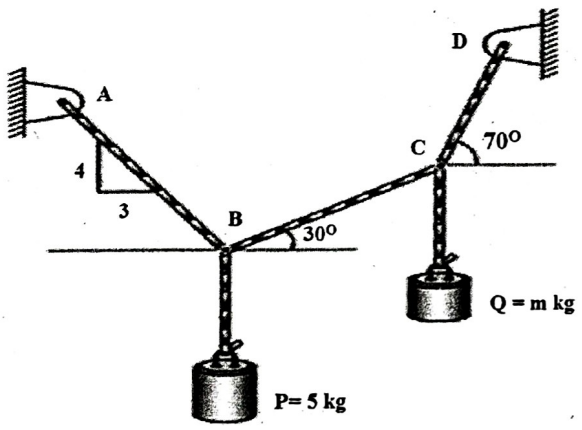
Q5. (10 Marks)

a. In the figure, a horizontal force of magnitude 890 N is acting on a particle at C which is attached to two strings connected to A and B respectively. Given that the particle is in equilibrium and the two strings and the particle all lie in the same vertical plane, find the tension in the two strings to the nearest newton. CO 2



OR

b. Determine the mass of block Q. CO 2



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